



United International University

B.Sc. in Data Science

Course Syllabus

1. **Course Title:** Object Oriented Programming for Data Science Laboratory
2. **Course Code:** DS 1116
3. **Trimester and Year:** Spring 2026
4. **Pre-requisites:** DS 1502.
5. **Credit Hours:** 1.0
6. **Section:** BA
7. **Class Hours:** Tue 8:30 - 11:00
8. **Class Room:** 627
9. **Instructor's Name:** Muhammad Anwarul Azim
10. **Email:** anwarul@cse.uiu.ac.bd
11. **Office:** 619(D)
12. **Counselling Hours:** See Ucam
13. **Text Book:**
 - (i) Python Crash Course, 3rd Edition: A Hands-On, Project-Based Introduction to Programming, Eric Matthes, No starch press.
14. **References:**
 - (i) Automate the Boring Stuff with Python, Al Sweigart
 - (ii) Python for Data Analysis, Wes McKinney
 - (iii) Fluent Python, Luciano Ramalho
 - (iv) Intro to Python for Computer Science and Data Science: Learning to Program with AI, Big Data and the Cloud, Deitel and Deitel.
 - (v) Learn Python the Hard Way: A Very Simple Introduction to the Terrifyingly Beautiful World of Computers and Code Book by Zed Shaw
 - (vi) <https://www.geeksforgeeks.org/python-oops-concepts/>
15. **Course Contents (approved by UGC):** Philosophy of Object Oriented Programming (OOP); Advantages of OOP over structured programming; Abstraction and Encapsulation, classes and objects, access specifiers, static and non-static members; Constructors, destructors, and copy constructors; Array of objects, object pointers, and object references; Inheritance: single and multiple inheritance; Polymorphism: overloading, abstract classes, virtual functions and overriding; Exceptions; Object Oriented I/O; Template functions and classes; Multithreaded Programming.

16. **Course Outcomes:**

- CO1** Apply the Object-Oriented Programming Features in solving programming problems.
CO2 Understand using modern/popular IDEs to develop OOP-based applications.
CO3 Develop OOP-based Data Science Project using standard framework-specific libraries.
CO4 Interpret programs written in a programming language used in data science.
CO5 Develop simple data science applications using a programming language used in data science.

17. **Teaching Methods:** Lecture, Demonstration, Projects, Q/A

18. **CO with Assessment Methods:**

CO	Assessment Method	(%)
-	Attendance	10
-	Class Evaluations	Best one (CT 1: 20, CT: 20)
-	Home Assignments	10
-	Project	Proposal: 5, Two Updates: (5+5), Final demo: 5 = 20
CO1,CO2,CO3	Midterm exam	20
CO3,CO4,CO5	Final exam	20

19. **Mapping of COs and Program outcomes:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	x							
CO2	x							
CO3		x						
CO4	x							
CO5			x					

20. **Lecture Outline:**

Class	Topic	CO	Reading Ref	Activities/Outcome
1	Overview of the course. Role of OOP in Data Science. Basic Review of the control structures and Concepts. Lab Project Deceleration	CO1	Matthes Ch 1-8, 9	Lecture, Demonstration,
2	Intro to OOP. Classes and Objects. Instance Variables, Scopes, Constructors, Project Proposal Draft.	CO1	Matthes Ch 9, McKinney 3.2	Lecture, Demonstration
3	More on Classes and Objects, Methods, Compositions, Encapsulation, Access Modifiers, Name Mangling, Proposal Finalization.	CO2	Ramalho Ch 8, 11	Lecture, Demonstration
4	Inheritance, Method Overriding, Abstraction, Lab CT 01.	CO2	Matthes Ch 9, Ramalho Ch 11	Lecture, Demonstration, Assessment.
5	Multiple Inheritance, Polymorphism, Operator Overloading and Dunder methods	CO2	Ramalho Ch 1, 9-10	Lecture, Demonstration
6	Lab Midterm	CO1, CO2, CO3		Assessment

Class	Topic	CO	Reading Ref	Activities/Outcome
7	GUI I: Introduction to Tkinter, widgets, layout managers, callback functions, event-driven programming Project Update I	CO4	Matthes Ch 16	Lecture, Demonstration
8	GUI II: Organizing GUI code, separation of logic from interface (MVC-lite), multi-file GUI structure, Home Assignment Deceleration	CO4, CO5	Matthes Ch 16	Lecture, Demonstration
9	Exception Handling, custom exceptions, File I/O (text/CSV/JSON), pathlib, context manager, serialization CT 2	CO2, CO3	Matthes Ch 10; Ramalho Ch 18	Lecture, Demonstration, Assessment
10	Testing and Packaging: use of assert, unittest basics, multi-file program structure, modules and packages, import rules, Project Update II	CO3	Matthes Ch 11; Ramalho Ch 21	Lecture, Package a small program and write tests
11	Functions as objects, wrapper, basic decorators, decorators on methods, iterator, generator, Project Judging	CO2, CO5	Matthes Ch 9; Ramalho Ch 9	Lecture, Demonstration
12	Lab Final	CO1, CO2, CO3, CO4, CO5		Assessment

Letter Grades

Letter Grade	Grade Point	Marks	Assessment
A	4.00	90-100	Outstanding
A-	3.67	86-89	Excellent
B+	3.33	82-85	Very Good
B	3.00	78-81	Good
B-	2.67	74-77	Above Average
C+	2.33	70-73	Average
C	2.00	66-69	Below Average
C-	1.67	62-65	Poor
D+	1.33	58-61	Very poor
D	1.00	55-57	Pass
F	0.00	<55	Fail
I	0.00	-	Incomplete
W	0.00	-	Withdraw
R	0.00		Repeat/Retake

Program Outcomes

Graduates of the program will have an ability to:

- PO1** Analyze a complex problem and to apply principles of data science and other relevant disciplines to identify solutions.
- PO2** Design, implement, and evaluate solutions of complex problems to meet a given set of requirements.
- PO3** Apply theory, techniques, and tools throughout the data science lifecycle and employ the resulting knowledge to satisfy stakeholders' needs.
- PO4** Communicate effectively in a variety of professional contexts.
- PO5** Recognize professional responsibilities and make informed judgments based on legal and ethical principles in practice.
- PO6** Function effectively as a member or leader of a team engaged in activities to solve a wide variety of infrequently encountered problems in diverse areas.
- PO7** Acquire and apply new knowledge as needed, using appropriate learning strategies.
- PO8** Demonstrate understanding of the skills required to be an entrepreneur and ability to apply entrepreneurial skills.